

PARTS LIST

Semiconductors:

- IC1 - 7812, 12V voltage regulator
- IC2 - BQ2450 charge controller
- IC3 - LM17300 comparator
- IC4 - UCC383 regulator
- IC5 - TPS2051C power switch
- IC6 - TPS73733 regulator
- IC7 - BQ32000 RTC
- IC8 - ATmega8 microcontroller
- IC9 - LM35 temperature sensor
- D1-D5 - 1N4007 rectifier diode
- D6 - 1N5422 Schottky diode
- D7 - 1N5822 Schottky diode

LED1, LED8,

- LED9 - 5mm LED
- LED2-LED7 - 1-watt white LED
- RGB1 - RGB LED (4-pin)
- T1 - BD240 pnp transistor
- T2-T5 - BC547 npn transistor

Resistors (all 1/4-watt, $\pm 5\%$ carbon, unless stated otherwise):

- R1 - 0.47-ohm, 1-watt
- R2, R6, R8, R9 - 1-mega-ohm
- R3, R25 - 470-kilo-ohm
- R4, R14, R23 - 47-kilo-ohm
- R5, R15 - 22-kilo-ohm
- R7 - 67-kilo-ohm
- R10 - 470-ohm
- R11, R22 - 100-ohm
- R12, R50 - 47-ohm
- R13 - 22-ohm
- R16, R19, R24, R44-R46 - 4.7-kilo-ohm
- R17, R20 - 2-kilo-ohm
- R18, R31, R43 - 10-kilo-ohm
- R21, R26 - 220-ohm
- R27 - 150-kilo-ohm
- R28 - 50-kilo-ohm
- R29, R30 - 2-ohm, 1-watt
- R32-R38 - 120-ohm
- R39-R42 - 1-kilo-ohm
- R47-R49 - 56-ohm

Capacitors:

- C1 - 1000 μ F, 25V electrolytic
- C2, C3, C6, C13 - 100nF ceramic disk
- C9, C10 - 10nF ceramic disk
- C4 - 220 μ F, 25V electrolytic
- C5 - 220 μ F, 16V electrolytic
- C7 - 100 μ F, 16V electrolytic
- C8, C11, C12 - 10 μ F, 10V electrolytic
- C14 - 1 μ F, 10V electrolytic

Miscellaneous:

- CON1, CON2 - 2-pin terminal connector
- CON3-CON7, CON9, CON13, CON16 - 2-pin connector male and female
- CON8 - USB A-type connector
- CON10, CON14 - 7-pin connector male and female
- CON11, CON15 - 4-pin connector male and female
- CON12 - 6-pin connector male and female
- PZ1 - Piezobuzzer
- RL1 - 12V, 100 PCB-mounted relay
- SP1 - 10-12V, 7W solar panel
- SI-S3 - On/off switch
- S4-S6 - Push-to-on tactile switch
- BATT1 - 6V, 4.5Ah rechargeable battery
- BATT2 - 3V cell (CR2032)
- XTAL1 - 32.768kHz crystal
- X1 - 230V AC primary to 12V, 1A secondary transformer
- DIS1-DIS4 - L7543 common cathode 7-segment display
- DIS5 - USB-to-micro USB cable for mobile charging
- DIS6 - 230V AC mains power supply

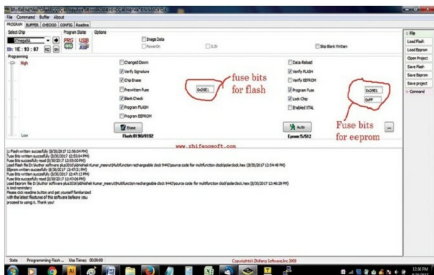


Fig. 7: Fuse bits

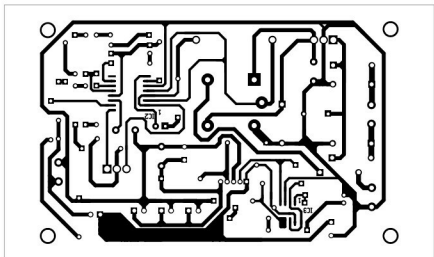


Fig. 8: Actual-size PCB layout of power unit

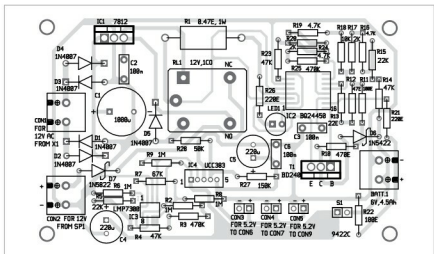


Fig. 9: Components layout for the PCB in Fig. 8